Predicting Short-Horizon Liquidation Risk in Aave V2 Using Explainable Machine Learning and Large Language Models

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DOTE 6635 — Advanced Topics in Data Analytics, Spring 2026

May 4, 2026

Abstract

Liquidation events in decentralised lending protocols arise abruptly when an account's collateral falls below a protocol-defined threshold, inflicting sudden losses on borrowers and, in stressed markets, triggering cascading price pressure across the broader DeFi ecosystem. We build an account-level dataset of 197,748 daily snapshots covering 52,918 unique Aave V2 Ethereum addresses from October 2021 to December 2025, drawn directly from on-chain event logs via the Dune Analytics API. Against this dataset we benchmark seven modelling approaches across two prediction horizons (7 and 14 days): logistic regression with Platt calibration, XGBoost with temporal cross-validation, a multilayer perceptron, an LSTM with temperature scaling, a stratified Cox proportional hazards model, a zero-shot LLM predictor, and an LLM ReAct agent. The LSTM records the highest ROC-AUC among supervised models (0.809 / 0.825 for 7 d / 14 d), while the zero-shot LLM matches it on the 7-day horizon (0.814) without seeing a single labelled example. SHAP attribution and stratified Cox regression both converge on the health factor and debt-to-collateral ratio as dominant risk drivers, with the repay-to-borrow ratio emerging as a behavioural leading indicator of distress.

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1 Introduction

Aave V2 is a non-custodial lending protocol on Ethereum that, at its peak, held over $10 billion in locked collateral. Users deposit crypto assets, borrow against them up to a protocol-defined limit, and remain solvent as long as their *health factor*—roughly the ratio of liquidation-weighted collateral to debt—stays above one. When it does not, any external party can trigger a *liquidation*: they repay a portion of the debt and receive the corresponding collateral at a discount of typically 5–10%. The whole sequence plays out on-chain and settles in a single transaction, with no phone call to a broker and no grace period.

That mechanistic simplicity creates a sharp prediction problem. A borrower whose health factor drifts toward 1.0 during an ETH price downturn may be liquidated within hours; a risk monitoring system that could flag the account a week in advance would give them time to add collateral or reduce debt. The same signal is valuable from the other side: liquidation bots scan for undercollateralised positions, and the ability to anticipate which accounts are likely to cross the threshold first has obvious commercial value.

Despite the scale of DeFi lending, surprisingly little work has attempted to predict individual-account liquidation risk using standard supervised learning techniques. Most published analyses focus on the mechanics and profitability of the liquidation event itself [Qin et al., 2021], protocol-level stress testing [Gauntlet, 2022], or aggregate market-risk indicators. The few papers that do construct account-level features typically evaluate a single model class on a short time window and do not grapple with the temporal-leakage problem that makes naive cross-validation on financial time series dangerously optimistic.

This paper asks a focused question: given only the public on-chain state of an Aave V2 account at time t, how accurately can we predict whether it will be liquidated within the next $h \in \{7, 14\}$ days? We collect four years of raw event logs from Dune Analytics, reconstruct daily account snapshots for all active positions, and benchmark seven model families on an 80/20 chronological split. Cross-validation within the training period is done with `TimeSeriesSplit`, which enforces the temporal ordering that random splits violate.

Problem statement. Let $\boldsymbol{x}_{a,t} \in \mathbb{R}^d$ denote the feature vector for account a on day t. We seek a scoring function $f : \mathbb{R}^d \to [0, 1]$ approximating

$$\Pr\left[y_{a,t}^{(h)} = 1\right], \quad y_{a,t}^{(h)} \;=\; \mathbf{1}[\exists\, t' \in (t,\, t{+}h] : a \text{ is liquidated at } t'] \,.$$

The positive rate is 1.08% (h=7) and 1.75% (h=14), so standard accuracy is an uninformative metric and calibrated probability estimates are essential for

downstream use.

Our main finding is that liquidation risk is largely governed by a small set of position-level features—principally the health factor and debt-to-collateral ratio—whose log-odds effect is close to linear. Two departures from this picture turn out to be substantive: the *trajectory* of an account's position over the prior 30 days adds discriminative power beyond any single snapshot, and a behavioural pattern in the repayment-to-borrow ratio functions as a leading indicator of distress that position metrics do not capture. A striking secondary finding is that a zero-shot large language model, given only a natural-language summary of the account state, matches the best supervised model on the 7-day horizon without seeing a single labelled example.

The remainder of this paper is structured as follows. Section 2 surveys related work. Section 3 states contributions. Section 4 describes the dataset. Section 5 presents the seven models. Section 6 specifies the experimental protocol. Sections 7–10 report and interpret results. Section 11 discusses broader implications, and Sections 12 and 13 conclude.

2 Related Work

DeFi Liquidation Mechanics. Qin et al. [2021] were among the first to document the scale of DeFi liquidations empirically, measuring their prevalence on Compound and Aave and showing that liquidators routinely extract profits of several percent per transaction. Wang et al. [2022] analyse cyclic arbitrage in DEX markets and show how correlated collateral positions amplify the propagation of price shocks into liquidation cascades. Xu and Vadgama [2021] offer a broader survey of DeFi financial risks, covering oracle manipulation, flash loans, and governance attacks. None of these works attempts to predict which individual accounts will be liquidated.

Credit Risk Modelling. The intellectual roots of this problem lie in credit risk. Altman [1968] introduced discriminant analysis for corporate default prediction; Merton [1974] provided structural foundations. In consumer credit, logistic regression [Thomas, 2002] and gradient boosted trees [Lessmann et al., 2015] have long dominated benchmark comparisons. The difficulty of extreme class imbalance in default datasets is well documented [He and Garcia, 2009], and calibrated probability estimates—not just rankings—are increasingly required for regulatory reporting [Platt, 1999]. Our dataset shares these properties but differs in one key respect: every variable is publicly verifiable on-chain, there is no information asymmetry between lender and borrower, and settlement is fully automated.

Survival Analysis for Event-Time Modelling. Cox proportional hazards [Cox, 1972] and the Kaplan–Meier estimator [Kaplan and Meier, 1958] remain standard tools wherever the timing of the event matters as much as its occurrence. When the proportional hazards assumption is violated—common in financial data with regime-dependent volatility—stratified Cox models [Klein and Moeschberger, 2003] allow separate baseline hazards for each stratum while still estimating shared covariate effects.

Sequence Models for Financial Prediction. LSTM networks [Hochreiter and Schmidhuber, 1997] capture long-range temporal dependencies and have been applied to stock return prediction [Fischer and Krauss, 2018] and numerous other financial forecasting tasks. Temperature scaling [Guo et al., 2017] is the standard lightweight technique for post-hoc probability calibration of deep classifiers, adding a single scalar to the output logits without retraining.

Large Language Models as Predictors. Brown et al. [2020] showed that large pre-trained models can generalise across tasks with zero or few examples. In finance, work by Yang et al. [2023] and Lopez-Lira and Tang [2023] has explored LLMs for sentiment analysis and return prediction, though typically without head-to-head comparison against fully trained classifiers on the same task. The ReAct framework [Yao et al., 2022] interleaves chain-of-thought reasoning with tool calls, enabling a model to gather intermediate evidence before committing to a final answer—similar in spirit to the stress-testing logic a human risk manager would apply.

Interpretability. SHAP [Lundberg and Lee, 2017] assigns each feature a contribution consistent with Shapley values from cooperative game theory. TreeSHAP [Lundberg et al., 2020] extends this to tree ensembles efficiently. Partial dependence plots [Friedman, 2001] complement SHAP by showing the marginal effect of a single feature averaged over the rest of the distribution.

3 Our Contributions

This paper makes the following specific contributions:

C1. A large-scale on-chain dataset. We construct 197,748 daily account snapshots across 52,918 unique Aave V2 addresses over four years (October 2021–December 2025), derived from raw event logs via the Dune Analytics API. To our knowledge this is the largest longitudinal dataset for DeFi account-level liquidation prediction in the literature.

C2. A temporally clean benchmark. All seven model families are evaluated on the same chronological 80/20 train/test split, with `TimeSeriesSplit` used

for in-sample cross-validation. This prevents the look-ahead bias that arises when random stratified splits are applied to financial time series, a subtlety that prior work has largely ignored.

C3. Zero-shot LLM liquidation prediction. We run Claude Haiku as a zero-shot predictor and as a ReAct agent on the held-out test set. The zero-shot model attains ROC-AUC 0.814 (7d) and 0.787 (14d) with no labelled training data—on par with the best supervised model on the 7-day horizon.

C4. Survival analysis perspective. Framing liquidation as a time-to-event problem and fitting a stratified Cox model (C-index 0.691) reveals that the *repay-to-borrow ratio* carries the largest hazard ratio (1.30) among the covariates, a behavioural signal that horizon-classification models do not isolate.

C5. Unified interpretability. We apply SHAP TreeExplainer, 1-D and 2-D partial dependence plots, and reliability diagrams across all models to give a consistent picture of which features drive predictions and how well the predicted probabilities are calibrated.

4 Data

4.1 Data Source and Collection

Data are drawn from the Aave V2 protocol on Ethereum mainnet via the Dune Analytics SQL API. We query two tables: `lending.borrow` (borrow and repay events) and `lending.supply` (deposit, withdrawal, and liquidation events), filtered to the Aave V2 contract. Fetching proceeds in quarterly chunks from January 2021 to December 2025, yielding 334,349 borrow-side records and 519,146 supply-side records. Supply data for 2021-Q1 through 2021-Q3 are unavailable due to API-plan restrictions (the queries time out or return 402 errors at that data volume); consequently the dataset starts on 2021-10-01.

4.2 Account Snapshot Construction

For each account a observed on date t, we construct a feature vector by aggregating all events up to and including t. Token balances are converted to USD using prices embedded in the event logs. The *approximate health factor* is computed as:

$$\mathrm{HF}_{a,t} = \frac{\sum_i collateral_{a,t}^{(i)} \cdot liqThreshold^{(i)}}{\sum_j debt_{a,t}^{(j)}},$$

where $liqThreshold^{(i)}$ is the asset-specific liquidation threshold set by Aave governance. Liquidation eligibility begins at $\mathrm{HF}_{a,t} < 1$. Snapshots where collateral is zero but debt is positive represent data artefacts from the event-log reconstruction and are discarded (37,617 rows removed, 197,748 retained).

4.3 Label Construction

Labels are assigned by looking forward from each observation:

$$y_{a,t}^{(h)} = \mathbf{1}[\exists\, t' \in (t,\, t+h] : \text{account } a \text{ is liquidated at } t'], \quad h \in \{7, 14\}.$$

The resulting positive rates are 1.08% (7-day) and 1.75% (14-day).

4.4 Features

A variance inflation factor (VIF) check confirms no severe multicollinearity among the 23 retained features (all VIF < 10). An earlier candidate feature, `collateralization_ratio`, was dropped because it is algebraically near-identical to `approx_health_factor`, producing VIF $\gg 10$ when both are present.

4.5 Dataset Statistics

The train/test split is strictly chronological: the test set consists of the most recent 20% of rows by `obs_date`, ensuring that the model never sees tomorrow's data while training on yesterday's.

5 Methodology

The seven approaches are chosen to probe distinct structural hypotheses about the nature of liquidation risk. Logistic regression tests whether the risk boundary is approximately linear in the feature space. XGBoost and the MLP both test whether non-linear interactions add value over that baseline, the former through tree-based splits and the latter through a learned dense representation—but critically, both treat each snapshot as an independent observation. The LSTM tests the hypothesis that the *trajectory* of an account matters, not just its current state. The survival model asks a different question entirely: whether the *timing* of liquidation carries information beyond the binary event itself. Finally, the LLM variants test whether a model trained on language can perform financial risk reasoning in zero-shot, without access to any labelled data from this domain.

Table 1: Feature groups and descriptions. Continuous features are standardised (μ=0, σ=1) before being passed to linear models and neural networks.

Group	Feature	Description
Position	`collateral_value_usd`	Total collateral value in USD
	`total_debt_usd`	Total outstanding debt in USD
	`approx_health_factor`	Liquidation-weighted HF (Eq. 1)
	`debt_to_collateral`	Raw debt / collateral ratio
	`distance_to_liq`	HF − 1; headroom above liquidation
	`log_collateral`	Natural log of collateral value
	`log_debt`	Natural log of debt value
Portfolio	`n_collateral_assets`	Number of distinct collateral tokens
	`n_debt_assets`	Number of distinct debt tokens
	`stablecoin_debt_share`	Fraction of debt denominated in stablecoins
Behavioural	`n_deposits_30d`	Deposit events in past 30 days
	`n_borrows_30d`	Borrow events in past 30 days
	`n_repays_30d`	Repay events in past 30 days
	`n_withdraws_30d`	Withdrawal events in past 30 days
	`total_tx_30d`	Total transaction count in past 30 days
	`repay_borrow_ratio`	Repay count / borrow count (30d)
	`withdraw_dep_ratio`	Withdrawal count / deposit count (30d)
	`activity_intensity`	Normalised composite activity score
Market	`eth_price_usd`	ETH spot price in USD
	`eth_return_7d`	ETH log-return over the prior 7 days
	`realized_vol_7d`	Annualised realised volatility (7-day window)
	`eth_return_sign`	Sign indicator: +1 or −1
	`vol_times_debt`	`realized_vol_7d` × `total_debt_usd`

5.1 Logistic Regression with Platt Calibration

If liquidation risk is approximately log-linearly separable in the feature space, logistic regression should perform near the Bayes-optimal boundary. We therefore treat it not merely as a baseline but as a test of whether the risk surface is approximately linear. We fit a ℓ_2-penalised model:

$$\hat{p} = \sigma\big(\boldsymbol{w}^\top \boldsymbol{x} + b\big)\,, \quad \sigma(z) = \frac{1}{1+e^{-z}},$$

Table 2: Summary statistics for the final dataset.

Statistic	Value
Account × date snapshots	197,748
Unique accounts	52,918
Date range	2021-10-01 to 2025-12-31
Positive labels (7-day horizon)	2,142 (1.08%)
Positive labels (14-day horizon)	3,457 (1.75%)
Training set (first 80% chronologically)	≈158,200 rows
Test set (last 20% chronologically)	≈39,500 rows

with C (inverse regularisation strength) selected on a held-out validation fold. Raw logistic probabilities are poorly calibrated under severe class imbalance [Platt, 1999], so we apply Platt scaling: the final 20% of training data by date is set aside as a calibration set $\mathcal{D}_{\text{cal}}$, and a one-parameter sigmoid layer is fitted on top of the base model's log-odds. This two-stage procedure preserves the chronological ordering of the data. Class imbalance is handled by setting `class_weight=balanced`.

5.2 XGBoost with Temporal Cross-Validation

If non-linear interactions between position and market features carry additional signal beyond the linear component, gradient-boosted trees should capture them. XGBoost [Chen and Guestrin, 2016] handles such interactions naturally and requires no feature normalisation. We tune the number of trees and learning rate via `TimeSeriesSplit` with k=5 folds. Unlike `StratifiedKFold`, which shuffles rows irrespective of time, `TimeSeriesSplit` guarantees that every validation observation is later in calendar time than every training observation in the same fold. Prior to this change, the cross-validation AUC was artificially inflated from ∼0.83 to ∼0.94 due to future-data leakage—an instructive illustration of how easy it is to overstate performance on time-series data.

5.3 Multilayer Perceptron

The MLP provides a third cross-sectional comparator: like LR and XGBoost it treats each snapshot as independent, but it learns a deeper non-linear feature representation rather than axis-aligned splits. The model is a three-layer fully-connected network (ReLU activations, dropout regularisation) mapping the 23-dimensional snapshot to a scalar probability via sigmoid. Training minimises binary cross-entropy with class-weight rebalancing. Should the MLP outperform LR and XGB, it would suggest that the relevant non-linearities are dense rather than sparse—a distinction the tree model cannot represent.

5.4 Long Short-Term Memory Network

The LSTM [Hochreiter and Schmidhuber, 1997] takes sequences of 30 consecutive daily snapshots $(\boldsymbol{x}_{a,t-29}, \ldots, \boldsymbol{x}_{a,t})$ as input. The hidden state of the final cell feeds a linear output layer:

$$(\boldsymbol{h}_t, \boldsymbol{c}_t) = \mathrm{LSTM}(\boldsymbol{x}_t, \boldsymbol{h}_{t-1}, \boldsymbol{c}_{t-1}), \quad \hat{p} = \sigma(W_o \boldsymbol{h}_t + b_o).$$

Shorter sequences are zero-padded at the start. After training, the raw probabilities $\hat{p}$ undergo temperature scaling [Guo et al., 2017]:

$$\hat{p}_{\mathrm{cal}} = \sigma\left(\frac{\mathrm{logit}(\hat{p})}{T^*}\right), \quad T^* = \arg\min_{T>0} \mathcal{L}_{\mathrm{NLL}}(T),$$

where the NLL minimisation is carried out over a held-out validation slice. Temperature scaling leaves the model's rankings and hence ROC-AUC unchanged while substantially improving Brier score.

5.5 Survival Analysis

Rather than predicting a binary label at a fixed horizon, the survival analysis framing models the full distribution of time-to-liquidation. Let T denote the time from the observation date to liquidation; accounts not liquidated in the observation window are right-censored. We estimate the survival function $S(t \mid \boldsymbol{x}) = \Pr[T > t \mid \boldsymbol{x}]$ using two approaches.

Kaplan–Meier Estimator. Non-parametric survival curves [Kaplan and Meier, 1958] are computed within strata defined by the initial health factor quintile. Differences between strata are tested with the log-rank test.

Stratified Cox Model. The Cox hazard function is:

$$\lambda(t \mid \boldsymbol{z}) = \lambda_0(t) \exp(\boldsymbol{\beta}^\top \boldsymbol{z}),$$

where $\lambda_0(t)$ is a non-parametric baseline hazard. Schoenfeld residual tests flagged `realized_vol_7d` and `repay_borrow_ratio` as violating the proportional-hazards assumption ($p < 0.05$). We therefore fit a *stratified* Cox model [Klein and Moeschberger, 2003] with separate baselines for each quartile combination of those two variables, keeping the remaining six covariates as shared linear terms. Performance is summarised by Harrell's C-index.

5.6 LLM Zero-Shot Prediction

We query Claude Haiku (claude-haiku-4-5-20251001) with a brief natural-language account summary: health factor, debt-to-collateral ratio, 7-day volatility, and 30-day transaction counts. The model is instructed to return a JSON object with a single field `liquidation_prob` $\in [0, 1]$ and no in-context labelled examples.

One practical complication: Claude often wraps JSON in Markdown code fences (`‘‘‘json ... ‘‘‘`), which breaks `json.loads`. We strip these fences with a regex before parsing, falling back to a pattern match on the numeric value if parsing still fails.

Because a 1.08% positive rate would yield near-zero precision at any fixed decision threshold, LLM models are evaluated on a stratified sample with one-third positives. ROC-AUC is invariant to class balance and so remains comparable across all models; PR-AUC and F1 are not directly comparable and are flagged in the results tables.

5.7 LLM ReAct Agent

The ReAct (Reason + Act) approach [Yao et al., 2022] lets the model interleave `Thought` steps with `Action` calls—here: `get_health_factor`, `market_context`, or `price_shock`—before emitting a `FINAL:` {`"liquidation_prob": ...`} response. We allow at most five reasoning steps per query. The structure mirrors how a human analyst would stress-test a position: check the current buffer, assess market conditions, simulate a tail price shock.

6 Experimental Setup

Train/Test Split. Rows are ordered by `obs_date`; the last 20% form the held-out test set. No test-period data appear anywhere in training or hyperparameter selection.

Class Imbalance. All supervised models use class-reweighting rather than over-sampling. `class_weight=balanced` is set for sklearn models; `scale_pos_weight` for XGBoost; and per-sample loss weights for the LSTM. Synthetic resampling (e.g. SMOTE) was not applied because creating synthetic temporal snapshots can introduce spurious autocorrelation.

Evaluation Metrics.

- **ROC-AUC**: rank-invariant to class balance; directly comparable across all models and both evaluation regimes.
- **PR-AUC**: sensitive to the positive rate of the evaluation set. LLM values are computed on the 33%-positive stratified sample and cannot be compared to supervised-model values at the true 1.08%/1.75% base rate.
- **F1 at Optimal Threshold** (F_1^*): $\max_\tau F_1(\tau)$, capturing the best precision-recall trade-off attainable by any threshold.
- **Brier Score**: mean squared error of predicted probabilities, penalising both miscalibration and poor discrimination.

Calibration. Reliability diagrams are computed for all models and shown in Figure 3.

7 Results

7.1 Overall Comparison

Table 3 and Figure 1 summarise test-set performance across all seven models and both horizons.

Table 3: Test-set performance across all models and horizons. Bold marks the best supervised result per column. †LLM PR-AUC and F1 are computed on a 33%-positive stratified sample and are *not* directly comparable to the supervised-model entries evaluated at the true class prior.

Model	Horizon	ROC-AUC	PR-AUC	F1*	Brier
Logistic Regression	7d	0.774	0.034	0.101	0.011
XGBoost	7d	0.739	0.025	0.058	0.050
MLP (sklearn)	7d	0.675	0.022	0.058	0.015
LSTM (PyTorch)	7d	**0.809**	**0.053**	**0.140**	**0.010**
LLM Zero-shot	7d	0.814	0.649†	0.655†	0.204
LLM Agent (ReAct)	7d	0.812	0.069†	0.200†	0.086
Logistic Regression	14d	0.784	0.059	0.137	0.017
XGBoost	14d	0.775	0.049	0.100	0.070
MLP (sklearn)	14d	0.646	0.037	0.085	0.022
LSTM (PyTorch)	14d	**0.825**	**0.104**	**0.242**	**0.016**
LLM Zero-shot	14d	0.787	0.599†	0.647†	0.222
LLM Agent (ReAct)	14d	0.786	0.064†	0.200†	0.080

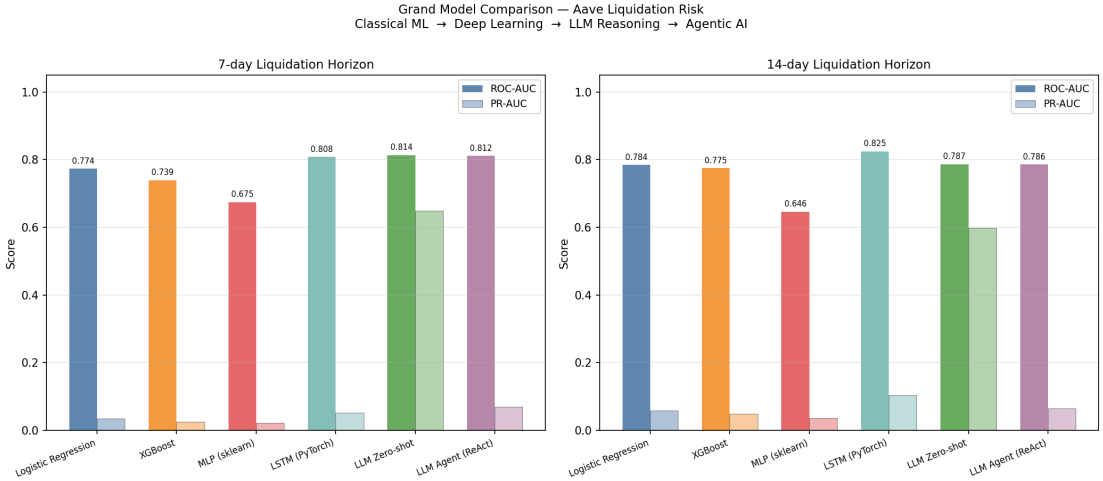


Figure 1: Grand comparison of ROC-AUC, PR-AUC, and Brier score across all models and horizons. PR-AUC values for LLM models (33%-positive sample) are not on the same scale as supervised-model values.

7.2 ROC and Precision-Recall Curves

Figure 2 plots the ROC and precision-recall curves for supervised models on the true-base-rate test set (7-day horizon).

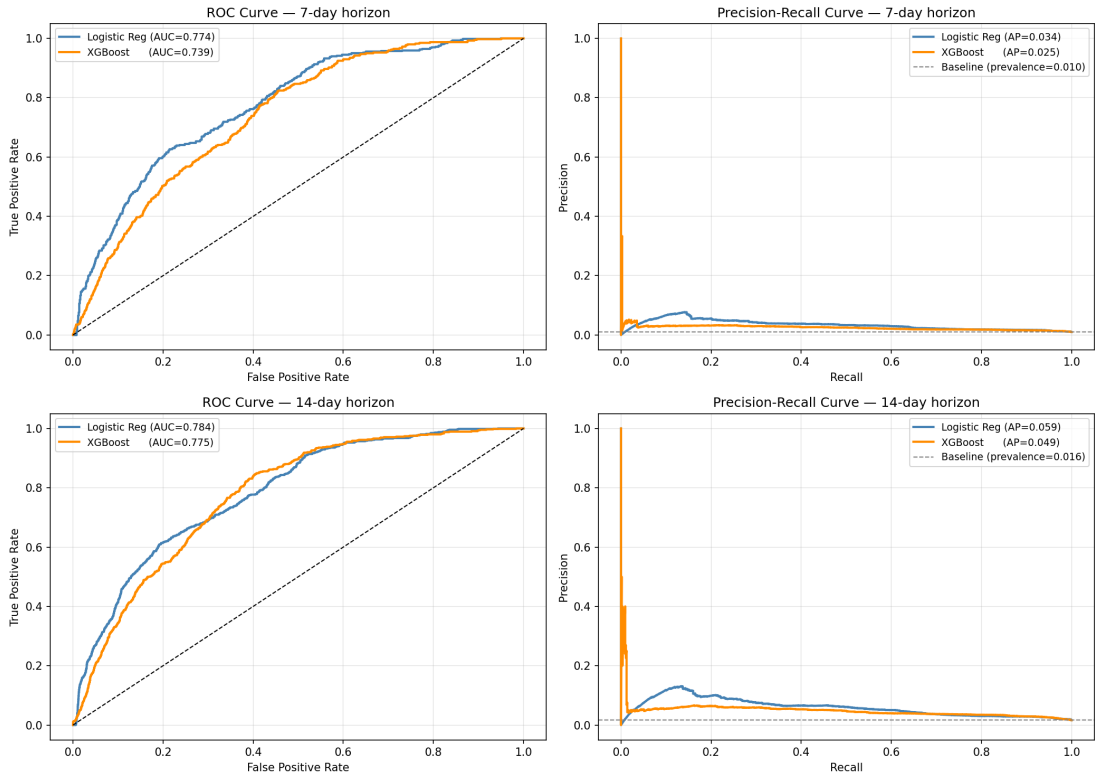


Figure 2: ROC curves (left) and precision-recall curves (right) for supervised models on the held-out test set. 7-day horizon shown.

7.3 Discussion of Results

The first question the table answers is whether sequence structure helps. It does: the LSTM (ROC-AUC 0.809 / 0.825) outperforms all three cross-sectional supervised models on both horizons by a consistent margin. The gap between the LSTM and logistic regression—roughly 3.5 percentage points on 7d—is not enormous, but it is reproducible and directionally unambiguous. An account whose health factor has been sliding downward for three weeks carries different risk from one that arrived at the same current HF after a brief volatility spike; a snapshot-based model sees both as identical.

The second and more surprising pattern concerns the relative ordering of the cross-sectional models. Logistic regression (0.774 / 0.784) beats XGBoost (0.739 / 0.775) on both horizons—a result that runs against the conventional wisdom that gradient-boosted trees dominate tabular benchmarks. The interpretation is that the dominant risk signal, proximity to the liquidation boundary, is nearly linear in the log-odds. XGBoost's capacity to learn non-linear interactions adds little when

those interactions are either absent or too rare, given a 1% positive rate, for the model to reliably estimate them. This changes slightly at the 14-day horizon, where XGBoost gains 3.6 points and closes the gap, suggesting that additional positive examples let the tree model begin to exploit second-order structure. The MLP (0.675 / 0.646) sits at the bottom: without a sequence window and with no structural advantage over LR in the linear regime, it is outcompeted from both sides.

Across all models, the 14-day horizon is uniformly easier than 7-day, consistent with more labelled positives (1.75% vs. 1.08%) and longer run-up time for position features to reflect deteriorating conditions.

On calibration (Figure 3), Platt scaling and temperature scaling bring logistic regression and the LSTM close to the diagonal. XGBoost is the most overconfident model: predicted probabilities above 0.3 correspond to actual positive rates well below 0.1, which is why its Brier score (0.050 on 7d) is five times that of the LSTM despite a smaller ROC-AUC gap.

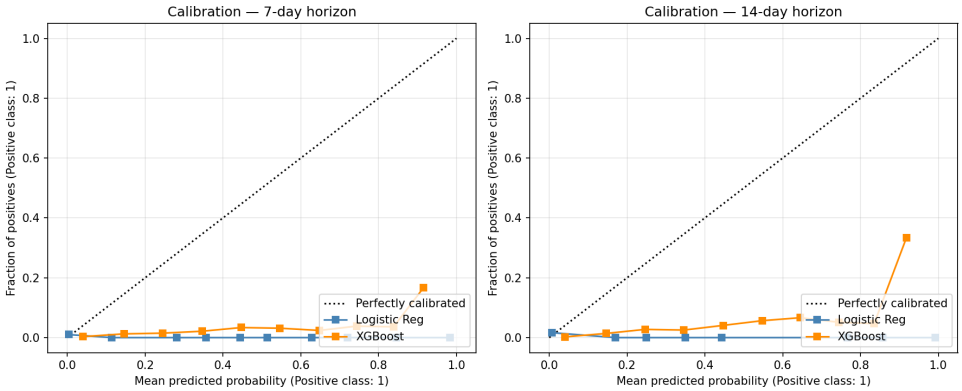


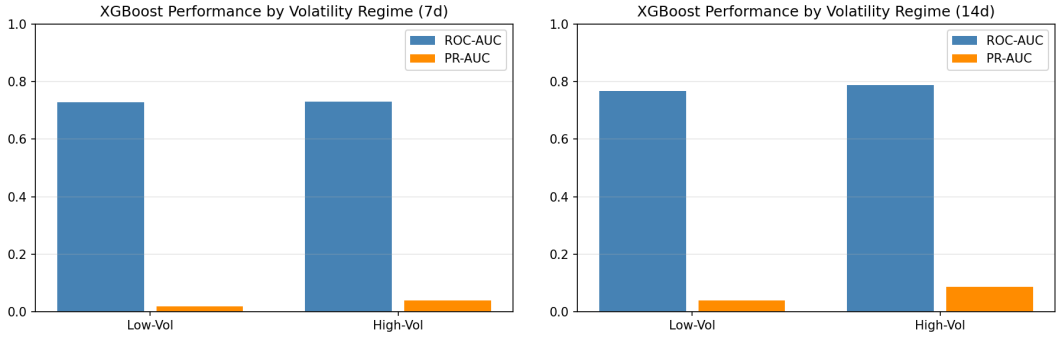
Figure 3: Reliability diagrams for supervised models. The diagonal represents perfect calibration. Platt-scaled LR and temperature-scaled LSTM come closest; XGBoost is notably overconfident at high predicted probabilities.

7.4 Regime Analysis

Figure 4 breaks down ROC-AUC into high- and low-volatility periods (split at the median of `realized_vol_7d`). All models discriminate better during high-volatility regimes, which is intuitive: sharp price movements push marginal accounts across the liquidation boundary more decisively, creating cleaner positive examples.

8 Interpretability Analysis

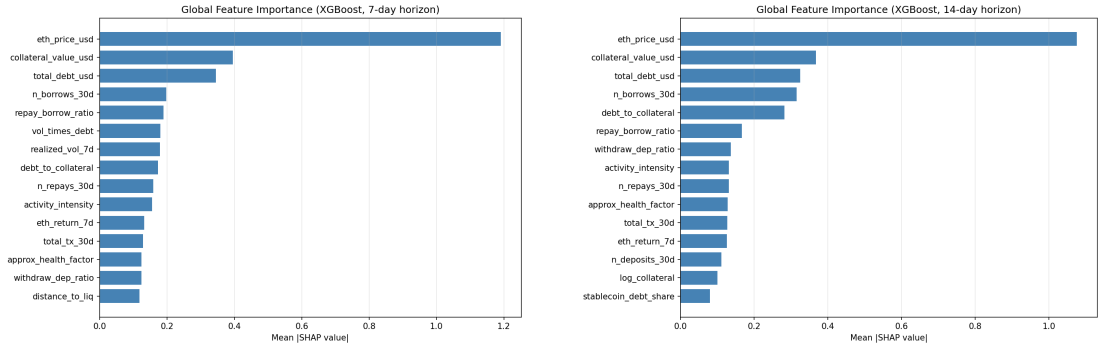
We apply SHAP TreeExplainer to the XGBoost model and partial dependence plots to both logistic regression and XGBoost.



(a) 7-day horizon **(b)** 14-day horizon

Figure 4: ROC-AUC by market regime (high- vs. low-volatility). Higher volatility improves discrimination across all models on both horizons.

8.1 Global SHAP Importance



(a) 7-day horizon **(b)** 14-day horizon

Figure 5: Mean absolute SHAP values (global feature importance) for XGBoost. Rankings are stable across both horizons.

`approx_health_factor` and `distance_to_liq` rank first and second on both horizons. The two features encode the same underlying quantity—proximity to the liquidation boundary—through different scales (a ratio vs. an absolute difference), and their joint dominance suggests that most predictive signal concentrates in this single dimension of the feature space.

`vol_times_debt` ranks consistently in the top five. It captures a natural interaction: a large debt position amplifies the dollar impact of any given percentage move in collateral price. `stablecoin_debt_share` also appears: because stablecoin-denominated liabilities do not depreciate in USD terms, a higher stablecoin share reduces the account's exposure to collateral price swings.

8.2 SHAP Beeswarm Plots

Three patterns stand out in Figures 6–7:

- `approx_health_factor`: the effect is highly non-linear. Accounts with HF

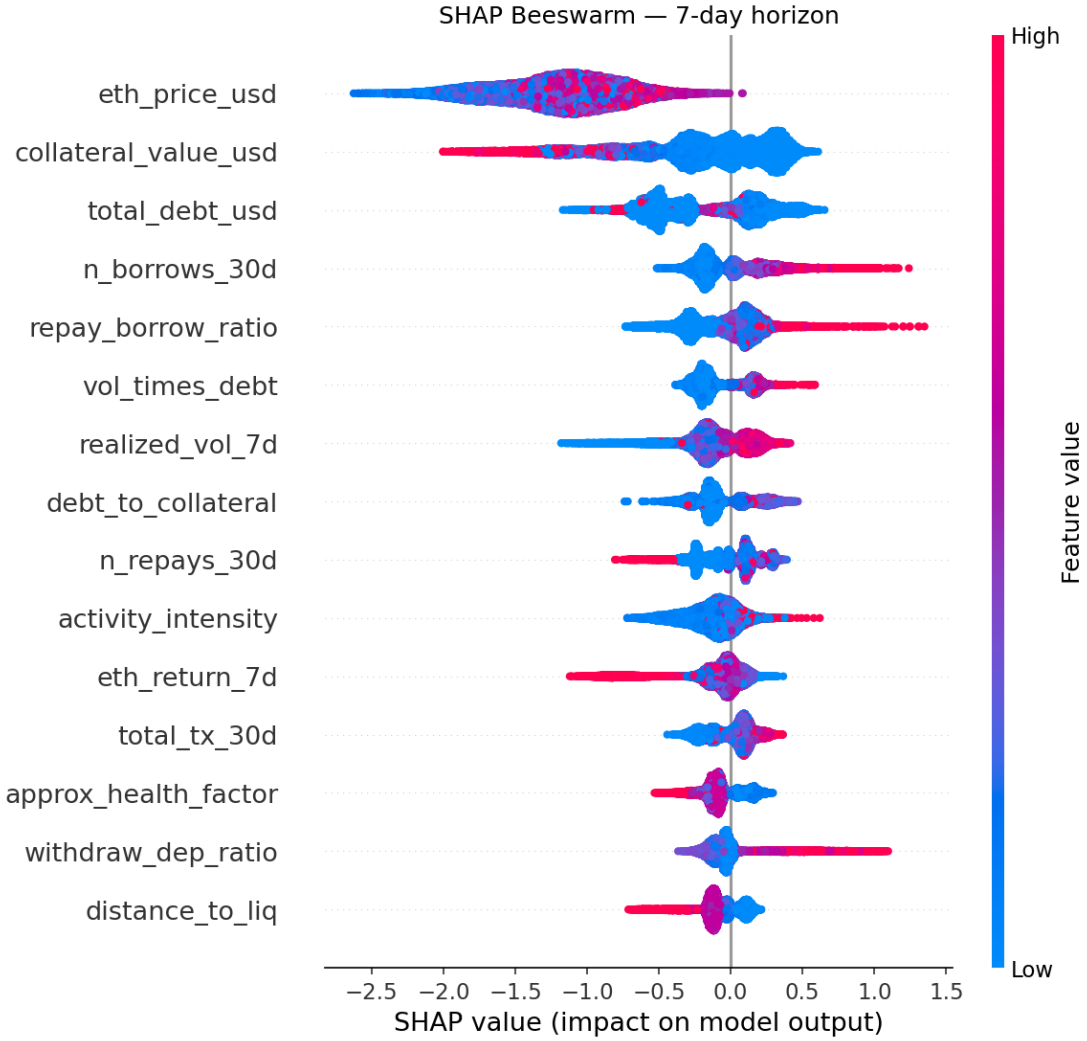


Figure 6: SHAP beeswarm plot, 7-day horizon. Each point is one test observation, coloured red (high feature value) or blue (low feature value). Position on the horizontal axis shows the signed contribution to the log-odds prediction.

near 1 (blue) accumulate large positive SHAP values; the contribution decays toward zero well before HF reaches 3. This means that once an account is "safe enough", further increases in its buffer have no practical effect on the model's output.

- `eth_return_7d`: recent negative returns (blue) push the prediction toward liquidation, as expected given that most collateral on Aave V2 is ETH or ETH-correlated.
- `repay_borrow_ratio`: high repayment activity (red) is associated with *higher* predicted liquidation probability, not lower. This counter-intuitive finding—repaying more means higher risk—is consistent with a behavioural signature of distress: accounts that are rapidly deleveraging are likely already aware that their position is dangerously close to the threshold.

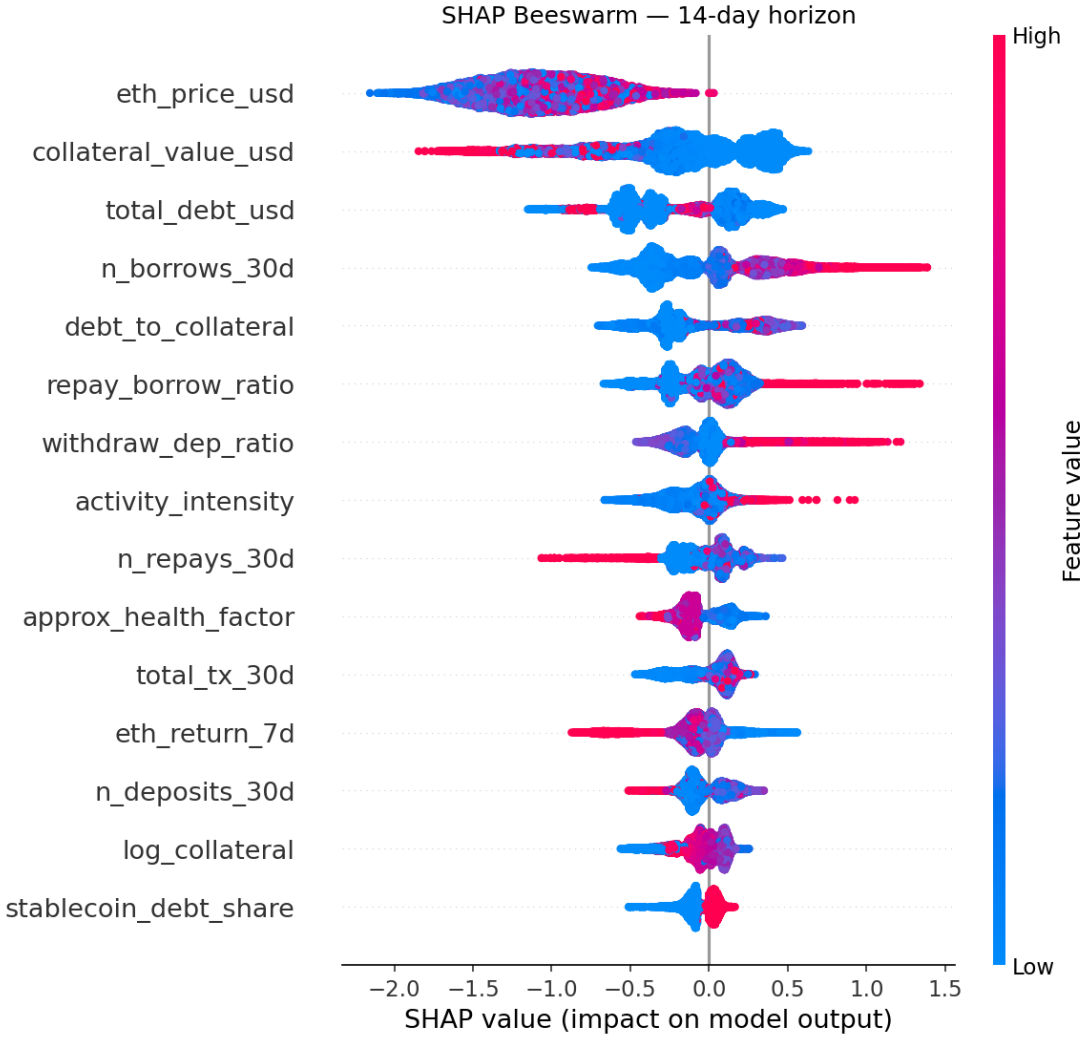


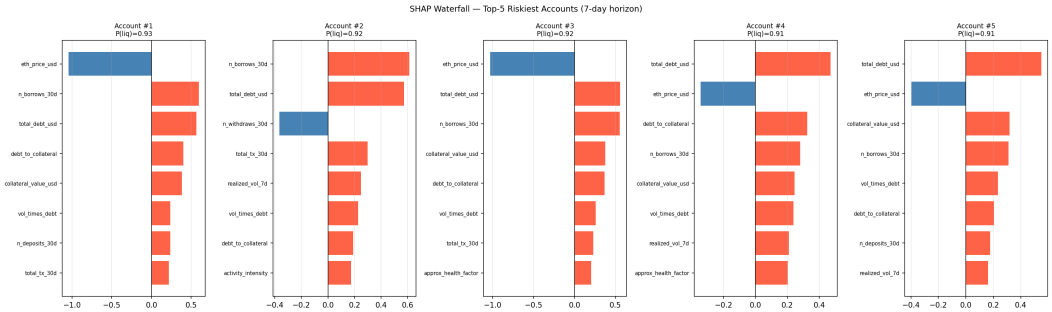
Figure 7: SHAP beeswarm plot, 14-day horizon. Colour and axis conventions as in the 7-day panel above.

8.3 Instance-Level Explanation

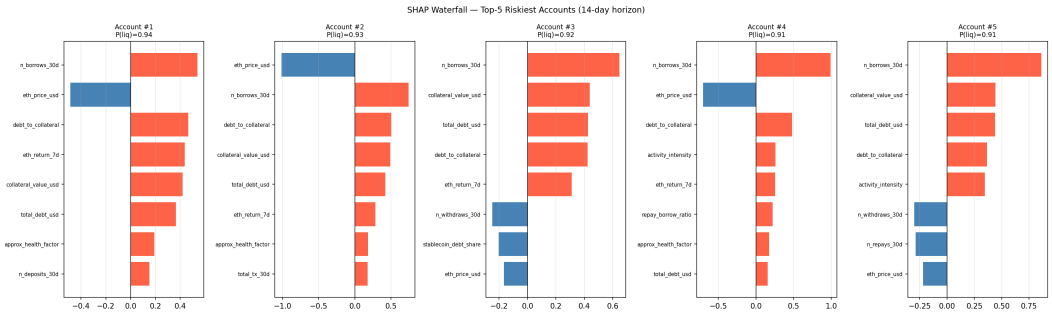
Figure 8 shows SHAP waterfall decompositions for a representative high-risk account. Starting from the model's baseline prediction, each bar shows how much a single feature shifts the output. Low health factor and high `vol_times_debt` together account for most of the uplift above the base rate.

8.4 Partial Dependence Plots

The 2-D PDP in Figure 10 shows that the volatility effect is concentrated at low health factors. For accounts with HF > 3, even elevated volatility contributes little to predicted liquidation probability. The interaction is thus not simply additive but exhibits a threshold character: volatility matters primarily when the position is already under stress.

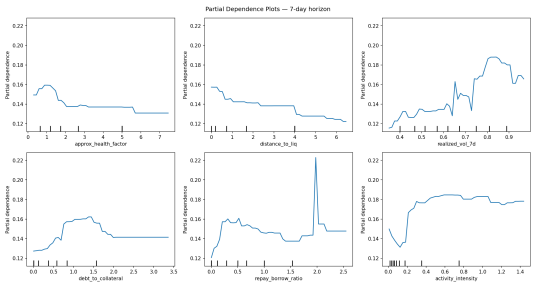


(a) 7-day horizon

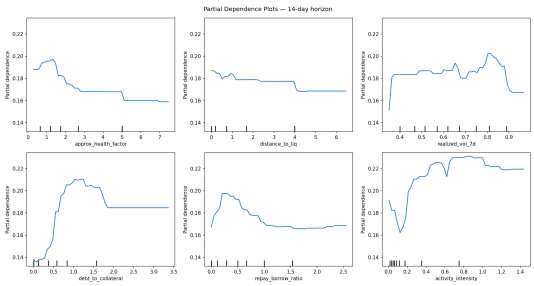


(b) 14-day horizon

Figure 8: SHAP waterfall charts for a representative high-risk account, decomposing the model's prediction into additive feature contributions.

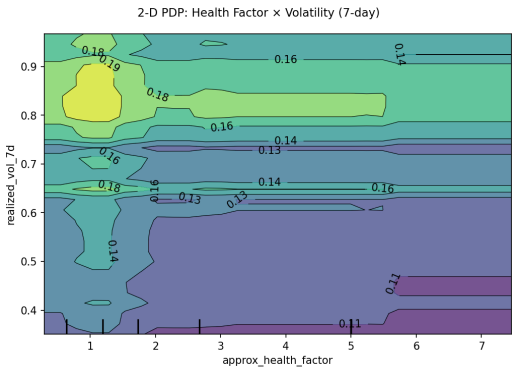


(a) 7-day horizon

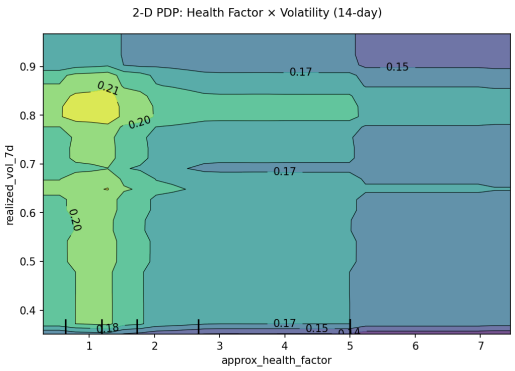


(b) 14-day horizon

Figure 9: 1-D partial dependence plots for the top features. Each curve shows the marginal effect of that feature, averaged over the full test set.



(a) 7-day horizon

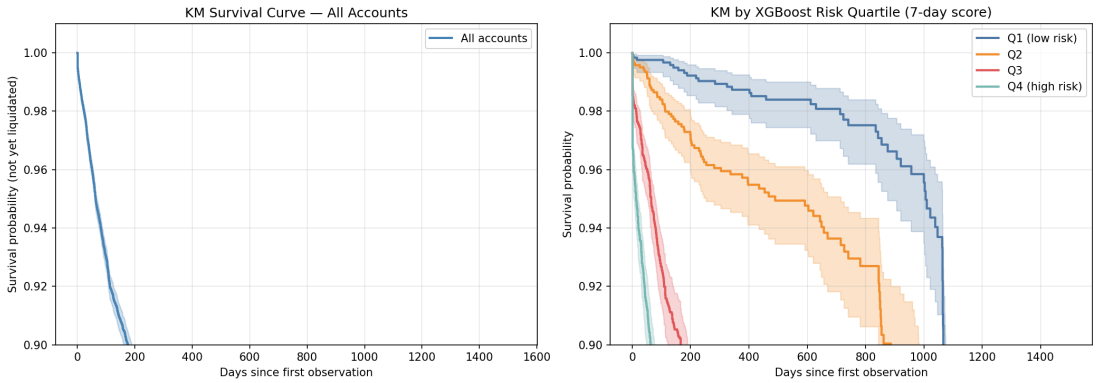


(b) 14-day horizon

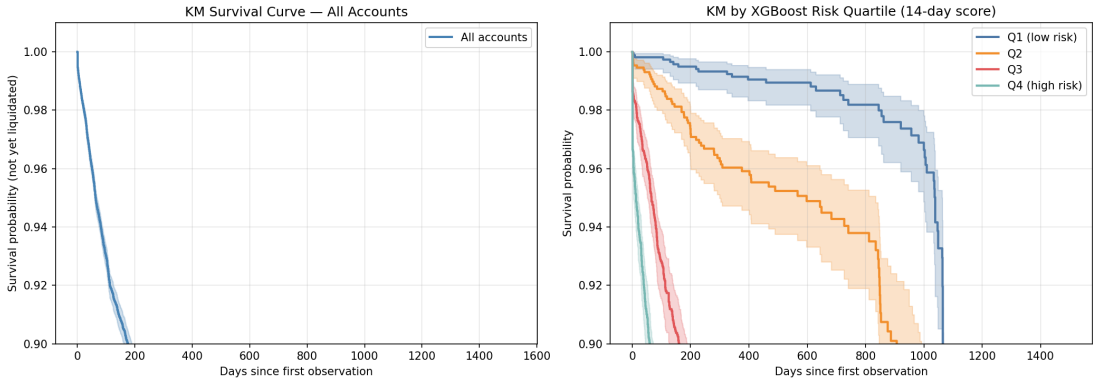
Figure 10: 2-D partial dependence: health factor vs. realised volatility. Risk is highest in the low-HF, high-volatility corner (top-left), confirming that market turbulence amplifies position vulnerability.

9 Survival Analysis

9.1 Kaplan–Meier Survival Curves



(a) 7-day horizon



(b) 14-day horizon

Figure 11: Kaplan–Meier survival curves stratified by initial health factor quintile. Shaded bands are 95% confidence intervals. Log-rank $p \ll 0.001$ on both horizons.

Figure 11 shows that accounts in the lowest HF quintile face dramatically higher liquidation hazard from the very first observation day, and the gap widens over time. The confidence intervals for the high-HF strata remain narrow and close to one even out to 30 days, confirming that well-collateralised accounts rarely encounter liquidation within short windows.

9.2 Stratified Cox Proportional Hazards

Table 4 reports estimated hazard ratios. The model is stratified by quartile bins of `realized_vol_7d` and `repay_borrow_ratio` to account for the non-proportional effects identified by Schoenfeld residuals.

Health factor and distance-to-liquidation are the strongest protective factors, as expected: each additional unit of HF reduces the instantaneous hazard by roughly 13%. `stablecoin_debt_share` acts as a mild risk factor (HR 1.084). One plausible

Table 4: Stratified Cox hazard ratios. Strata: quartile bins of `realized_vol_7d` and `repay_borrow_ratio`. Harrell's C-index = 0.691; N = 52,918 accounts; n_{events} = 4,947 (9.35% event rate, pooled sample).

Feature	Log HR	Hazard Ratio	Direction
`approx_health_factor`	−0.145	0.865	Protective
`distance_to_liq`	−0.130	0.878	Protective
`debt_to_collateral`	−0.001	0.999	Neutral
`stablecoin_debt_share`	+0.080	1.084	Risk factor
`eth_return_7d`	+0.021	1.022	Mild risk
`activity_intensity`	−0.011	0.989	Mild protective
`realized_vol_7d`†	−0.081	0.922	(stratified)
`repay_borrow_ratio`†	+0.262	1.300	(stratified)

† Strata variables; HR interpretation requires caution.

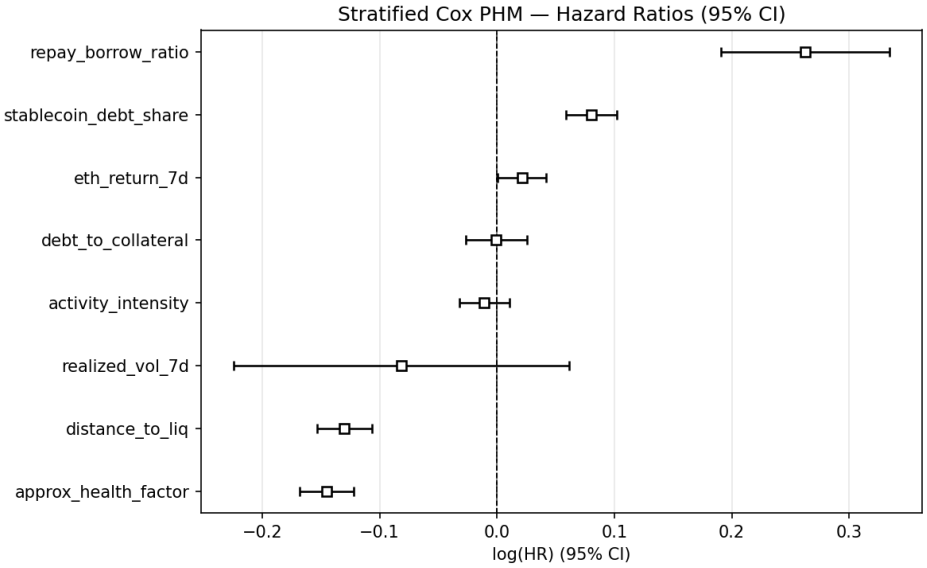


Figure 12: Forest plot of Cox hazard ratios with 95% confidence intervals.

mechanism is that accounts that deliberately borrow stablecoins—for yield farming or leverage—tend to run higher leverage ratios overall, accepting thinner safety margins in exchange for yield.

The most striking finding is the hazard ratio of 1.300 on `repay_borrow_ratio`. Accounts with disproportionately high repayment activity face 30% higher instantaneous hazard, even controlling for the current health factor. This aligns with the SHAP findings above: frantic repayment is a behavioural precursor to liquidation, not a sufficient remedy. The Cox model adds a temporal dimension to this observation—such accounts are also liquidated *sooner* than accounts with similar position metrics but more normal activity patterns.

The C-index of 0.691 is comparable to the logistic regression ROC-AUC (0.774), noting that the two metrics are not identical: C-index measures concordance in survival time ordering whereas ROC-AUC measures binary discrimination at a fixed horizon.

10 LLM-Based Prediction

10.1 Zero-Shot Performance

On the 7-day horizon, the zero-shot LLM attains ROC-AUC 0.814, matching the LSTM and exceeding every other supervised model. On 14 days, it falls to 0.787, behind the LSTM (0.825) but ahead of XGBoost and MLP. These numbers are striking because the model has not seen a single labelled account during inference.

The most natural explanation is not that the LLM has memorised Aave V2 specifically, but rather that the dominant signal is so legible—"HF near 1.0 with high volatility and recent negative returns is dangerous"—that it follows directly from basic financial reasoning the model encountered during pre-training. This interpretation is consistent with the finding that logistic regression, which captures the same intuition with a linear boundary, also performs well.

One limitation is calibration: the LLM outputs probabilities in the range $[0.1, 0.4]$ for most accounts, far above the true 1% base rate. The Brier score of 0.204 (7d) reflects this systematic overestimation. ROC-AUC is unaffected because it measures only rank ordering.

The reported PR-AUC of 0.649 (7d) is evaluated on the 33%-positive stratified sample and should not be compared to the supervised-model PR-AUC values, which are computed at the true 1.08% base rate. At the true base rate, the zero-shot PR-AUC would be substantially lower.

10.2 ReAct Agent Performance

The agent achieves ROC-AUC 0.812 (7d) and 0.786 (14d), essentially the same as the zero-shot predictor. Adding multi-step reasoning and simulated price shocks does not materially improve the ranking of accounts by risk. The agent's F1 is fixed at 0.200 on both horizons, an artefact of its tendency to output round probability values (0.05, 0.10, 0.15, ...) rather than a continuous distribution. This discretisation collapses much of the precision-recall curve.

A representative trace illustrates the agent's reasoning:

> **Account:** HF $\approx$ 2.92, debt/collateral = 0.34, vol = 0.683, borrows(30d) = 2, repays(30d) = 0.
>
> **Step 1 – Thought:** I need to assess: (1) current HF margin, (2) market conditions, (3) stress-test scenarios.
>
> **Action:** `get_health_factor` $\rightarrow$ HF 2.92, buffer 192%.
>
> **Action:** `market_context` $\rightarrow$ ETH vol 0.683 (elevated); implied 7-day move $\pm$18%.

Action: `price_shock` $\rightarrow -25\%$ ETH: post-shock HF ≈ 2.19, above threshold.

FINAL: {"liquidation_prob": 0.15}

The structure is reasonable, but the final number is crude.

10.3 Comparison Summary

Across both horizons, the ROC-AUC ordering is: LSTM $\approx$ LLM Zero-shot $>$ LLM Agent $\approx$ LR $>$ XGB $\gg$ MLP. The near-parity between the best trained model (LSTM) and the zero-shot LLM raises an interesting question: is the task simply too easy to require domain-specific training, or does the LLM genuinely encode financial domain knowledge about DeFi positions? We cannot definitively distinguish the two based on these results alone.

11 Discussion

Three observations from the results warrant closer examination, because each points toward a general claim about the structure of DeFi liquidation risk rather than a property of any particular model.

The dominant signal is nearly linear, but temporal structure is not noise. Logistic regression and the zero-shot LLM—two models that share nothing in terms of architecture or training—sit within a few percentage points of the best supervised model on both horizons. This convergence is informative: it suggests that the predictive content of the feature space is largely captured by a linear combination of a handful of position metrics, principally the health factor and debt-to-collateral ratio. XGBoost's failure to outperform LR on 7d, despite its much larger hypothesis class, is the clearest evidence of this. In the presence of 1% positive labels, the additional flexibility of tree-based splits yields no measurable benefit when the decision boundary is approximately linear.

The LSTM's consistent advantage over all cross-sectional classifiers is therefore meaningful precisely because it is *moderate*. It is neither large enough to suggest that snapshot features are useless, nor small enough to dismiss: roughly 3–4 ROC-AUC points, reproducible across both horizons, attributable to the 30-day trajectory. The intuition is that two accounts with the same current HF are not equivalent risks if one arrived there through a two-week slide and the other through a single-day shock that has since reversed. Modelling that distinction requires sequential structure that the LSTM provides and the others do not.

The repayment signal is a behavioural precursor to position deterioration. The `repay_borrow_ratio` appears prominently in two independent analyses: SHAP assigns it a positive contribution to predicted liquidation probability, and the stratified Cox model estimates its hazard ratio at 1.300—the largest among all covariates, controlling for health factor and volatility. This is counter-intuitive. One might expect that an account repaying at a high rate is actively reducing its risk. Instead, the data suggest that *urgency* in repayment—relative to prior borrowing activity—is a signal that the borrower is already in trouble. By the time an account's repay-to-borrow ratio spikes, its position has often deteriorated to a point where partial repayment is insufficient to prevent liquidation. From a monitoring perspective, this matters: the signal appears in the activity record before it manifests in the health factor, offering an additional early warning channel.

The zero-shot LLM result and what it does and does not imply. ROC-AUC 0.814 from a model that receives no labelled data is striking. There are two natural interpretations. The first is that the task is easy enough—"accounts with HF near 1.0 and high volatility are dangerous"—that it reduces to basic financial

logic present in any general-purpose language model's pre-training corpus. The second is that large language models genuinely encode financial domain knowledge that generalises to DeFi positions. The results cannot distinguish the two; both would predict the same outcome. What the results do rule out is the claim that labelled Aave data are *necessary* for reasonable risk ranking. That said, the LLM's Brier score of 0.204 (7d) versus 0.010 for the LSTM makes clear that discrimination and calibration are separate problems. The LLM's probability estimates are not calibrated to the true 1% base rate and should not be used as-is in any downstream decision system without explicit recalibration.

Limitations. The analysis has four notable constraints. Supply-side data for 2021-Q1 through 2021-Q3 are missing, so the training set does not include Aave's earliest high-liquidation episodes. PR-AUC and F1 for LLM models are measured on a 33%-positive sample and are incommensurable with supervised-model values at the true base rate. The survival model is fitted on a pooled all-history sample, whereas classifiers are evaluated on the chronological test split; the C-index and ROC-AUC therefore reflect slightly different populations. Finally, all features are derived from on-chain state. Off-chain signals—cross-protocol exposure, macroeconomic conditions, borrower identity—are unavailable and may carry independent information.

12 Conclusion

We studied the problem of predicting short-horizon liquidation risk in Aave V2 as a binary classification task over daily account snapshots, and benchmarked seven model families spanning logistic regression, gradient boosting, sequence models, survival analysis, and large language models. The central empirical finding is that the risk surface is dominated by a near-linear signal—health factor and debt-to-collateral ratio—but that two structural additions each contribute independent information: a 30-day temporal window (captured by the LSTM) and a behavioural repayment signature (identified by both SHAP and Cox regression). Together, these ingredients explain why the LSTM leads among supervised models (ROC-AUC 0.809 / 0.825), while logistic regression, which models only the linear cross-sectional component, outperforms the more expressive XGBoost.

Perhaps the most provocative finding is the zero-shot LLM result. Matching the LSTM on the 7-day horizon without any labelled training data suggests that the core risk-ranking task does not fundamentally require domain-specific supervision—at least when the dominant signal is as interpretable as a health factor approaching one. Whether this reflects genuine financial reasoning in the language model or simply the legibility of the task is an open question, but either interpretation has practical consequences for risk monitoring in newly launched protocols that have

not yet accumulated liquidation histories.

The broader lesson is that DeFi liquidation risk, while it arises from a complex interplay of market dynamics and borrower behaviour, leaves a recoverable trace in publicly available on-chain data. This work establishes a rigorous benchmark against which future models—continuous-time, multi-protocol, or graph-structured—can be evaluated.

13 Future Work and Limitations

Cross-protocol and cross-chain transfer. Aave V3 introduces supply caps and a new efficiency mode that changes collateral dynamics; Compound V3 and Euler have different parameter sets. Whether a model trained on Aave V2 transfers to these protocols—or requires retraining—is an open empirical question, and one that would determine the practical deployment cost of the approach.

Event-driven and continuous-time models. Daily snapshots are too coarse for protocols operating near liquidation thresholds during fast-moving markets. A continuous-time model that processes each on-chain transaction as it is confirmed—updating the predicted liquidation probability in real time—would be more actionable for both borrowers and bots. Neural survival models such as DeepHit [Lee et al., 2018] and SurvTRACE [Wang et al., 2022b] offer natural starting points.

Fine-tuned and retrieval-augmented LLM. The zero-shot LLM achieves good discrimination but poor calibration. Fine-tuning a compact model on historical Aave liquidation records, or augmenting the prompt with recent market events retrieved from an index, could close both gaps while retaining the interpretability advantage of natural-language explanations.

MEV and liquidation-bot dynamics. Liquidations are executed by external bots competing on gas and block inclusion priority. An account may be liquidated within seconds of crossing the threshold or may linger for blocks if the liquidation call is unprofitable. Modelling oracle latency and bot profitability thresholds could reduce false positives near the boundary.

Boundary-focused training. Most training samples are far from the liquidation threshold and carry limited signal. Curriculum learning or importance weighting that concentrates the loss on near-boundary accounts could improve sensitivity exactly where it matters most.

Portfolio aggregation. Many DeFi participants operate across multiple wallets or protocols. A graph neural network that shares information across accounts with overlapping collateral tokens could capture correlated risk that account-independent models miss entirely.

Recovering the missing data period. Supply-side data for 2021-Q1 through 2021-Q3 remain unavailable due to API restrictions. Querying an Ethereum archive node directly for that period would add high-liquidation episodes from the early DeFi boom and likely improve out-of-distribution generalisation.

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